

HITESH CHAUDHARI

+91 8929074747 hitesh.chaudhari2024@gmail.com linkedin.com/in/hiteshyc github.com/Hiteshyc

EDUCATION

Vellore Institute of Technology, Vellore

B.Tech in Computer Science and Engineering

2023 – 2027

CGPA: 8.89 / 10

TECHNICAL SKILLS

Languages: C, C++, Java, Python, HTML, CSS, JavaScript

Databases & Tools: MySQL, Git, GitHub, VS Code, Android Studio, Arduino IDE, MeshLab

Frameworks & Libraries: React, Pandas, NumPy, Scikit-learn, TensorFlow

EXPERIENCE

Samsung PRISM Research Project | *Agentic RAG Framework, LLMs, RAG, Vector Databases*

May 2026 – Present

- Developing an Agentic RAG framework for query decomposition and routing across multiple domain-specific knowledge bases.
- Researching semantic routing, vector databases, and low-latency retrieval techniques to improve retrieval accuracy and response time for Samsung device-support use cases.

PROJECTS

Quantitative Time-Series Modeling Engine | *Python, Pandas, NumPy, Plotly, Groww API*

April 2026

- Engineered a **multi-factor “Value Cost Averaging” model** and an **event-driven backtesting framework** to simulate performance against baseline models (DCA, Lump Sum).
- Constructed an automated data pipeline to acquire, clean, and process over **1,700 unique time-series datasets** (6+ months of 1-hour data for 218 listed companies) via the Groww API.
- Engineered **5+ factor feature set** (Wilder's RSI, ATR, rolling quantiles) modeling market momentum, volatility, & value; validated alpha via ROI aggregation and **hyperparameter tuning**.

Real-Time Public Health Outbreak Surveillance System | *MongoDB, Express.js, React, Node.js*

October 2025

- Built a MERN-based platform for doctor reporting, symptom tracking, and regional outbreak monitoring with JWT-secured access.
- Deployed automated outbreak detection using sliding-window threshold logic, generating alerts from **real-time patient reports**.
- Refined MongoDB schemas, indexes, and REST APIs, reducing latency and enabling **faster, region-wide outbreak detection**; created interactive dashboards for actionable insights.

Customer Feedback Intelligence System for Car Rentals | *Python, Streamlit, IBM Watson NLU*

July 2025

- Engineered an AI sentiment engine with IBM Watson NLU and a Streamlit dashboard to analyze 20–50 reviews per batch, providing 3-level sentiment scoring and concise issue summaries.
- Implemented automated detection of **10+ recurring issues** (delays, pricing, vehicle condition, staff behavior); generated downloadable **DOCX reports** reducing manual review effort by **~70%**.

RESEARCH PAPER

DSFFN: Dual-Stream Forgery Fusion Network for Cross-Domain Deepfake Detection | *Python, PyTorch*

April 2026

- Developed a dual-stream deepfake detector integrating spatial (RGB) and frequency (phase) cues to enhance robustness under domain shift; achieved an AUC of **98.01%** with a **5.05%** smaller cross-domain performance drop than the spatial-only baseline. *(draft submitted)*

CERTIFICATIONS

Gen AI using IBM WatsonX | IBM Career Education Program

June 2025

- Generative AI, NLP, Prompt Engineering, IBM WatsonX

ACTIVITIES

IEEE Professional Communication Society (IEEE-PCS), VIT Vellore

April 2025 – February 2026

General Secretary

- Directed recruitment, increasing member engagement by **40%** and fostering a collaborative environment.
- Facilitated **10+ technical workshops, 5+ speaker sessions**, and multiple networking events, benefiting **500+ students**.
- Supervised outreach and marketing, boosting event participation by **60%** through strategic promotions.